

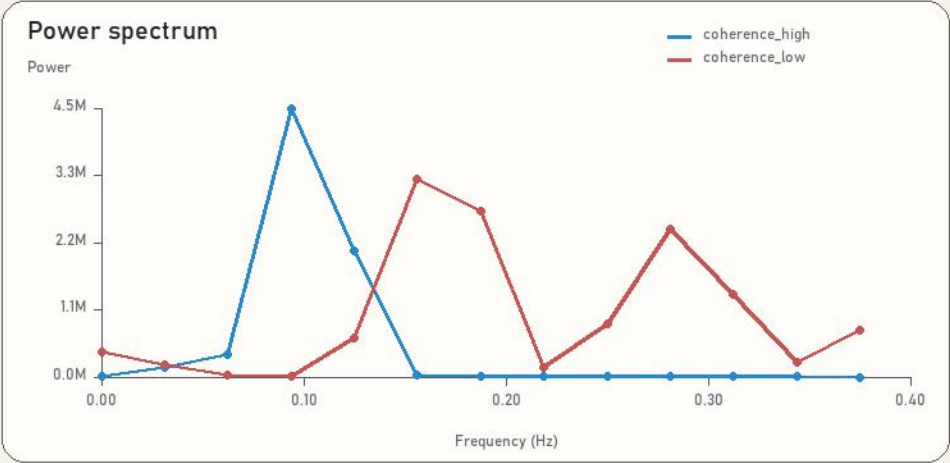
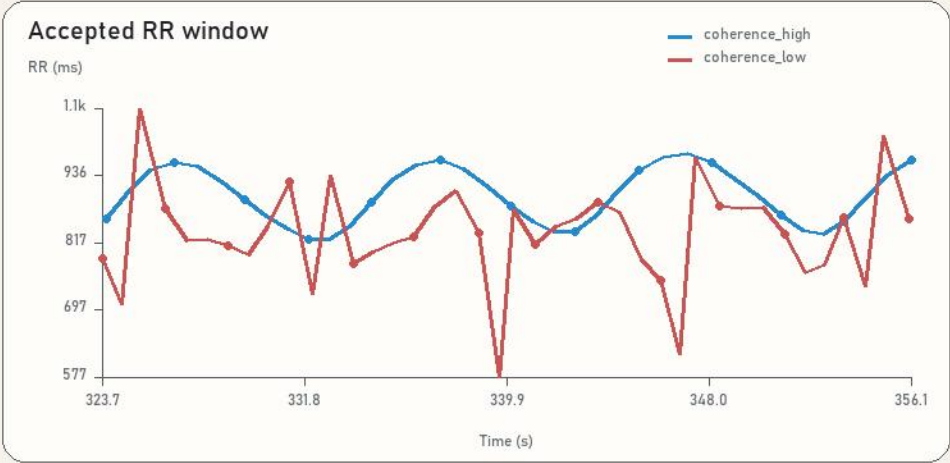
Synthetic showcase matrix

Deterministic scenarios exported from SyntheticBio using showcase-v1.

Scenario summary					
Scenario	Coherence	RMSSD	Int SampEn	Amp SampEn	State
breathing_pause	0.310	9.6	0.569	0.349	4
coherence_high	0.517	27.5	0.000	--	4
coherence_low	0.286	130.8	0.991	1.099	4
entropy_high	0.300	13.8	1.230	0.426	4
entropy_low	0.327	8.5	0.044	0.148	4
entropy_rising	0.324	13.3	0.890	0.445	4
flat_breathing	0.268	3.3	--	--	2
hrv_high	0.107	58.5	0.514	0.938	4
hrv_low	0.335	3.0	0.015	0.189	4
irregular_rr	0.200	186.0	1.429	0.301	4
jittered_breathing	0.257	13.3	1.174	0.426	4
off_10bpm	0.502	22.8	0.000	1.303	4
off_12bpm	0.445	17.3	0.625	0.393	4
off_18bpm	0.181	11.6	0.537	1.217	4
off_24bpm	0.265	6.6	0.379	0.823	4
regular	0.322	9.7	0.190	0.422	4
resonance_010hz	0.516	27.5	0.000	3.135	4

Coherence derivation

Raw RR intervals feed the spline-resampled tachogram and PSD solve used for the normalized coherence score.

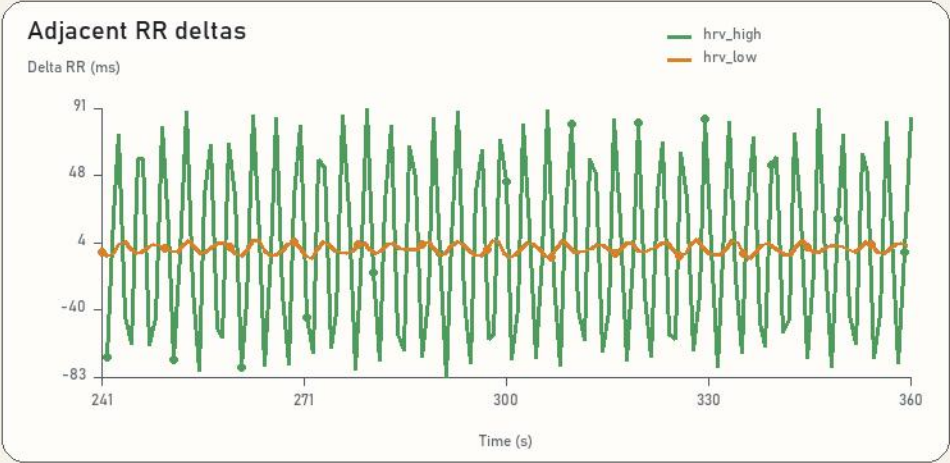
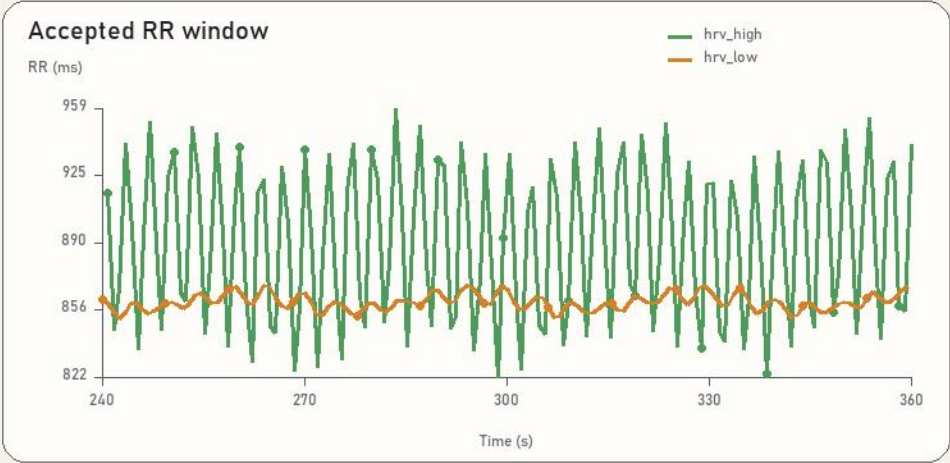


Peak-window summary				
Scenario	Peak freq	Peak power	Total power	Norm score
High	0.097 Hz	124216.950	224684.170	0.517
Low	0.167 Hz	102350.010	434405.300	0.286

Normalized coherence = P_peak / P_total; the SVG/PNG pair is backed by analysis.json and the formula sheet uses the same runtime telemetry.

HRV derivation

Accepted RR intervals produce adjacent RR deltas, which then determine RMSSD, SDNN, pNN50, and lnRMSSD.

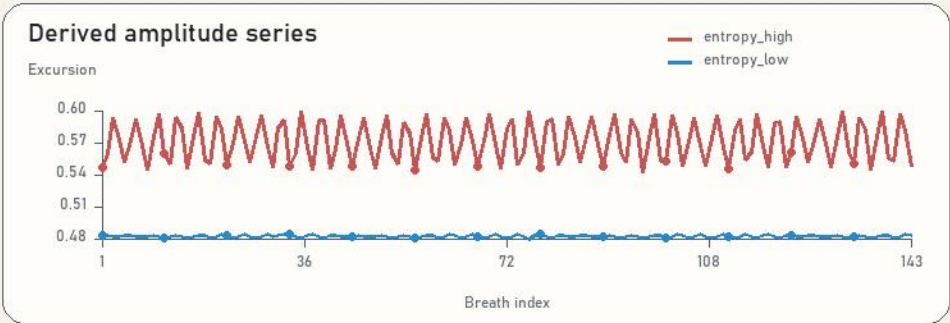
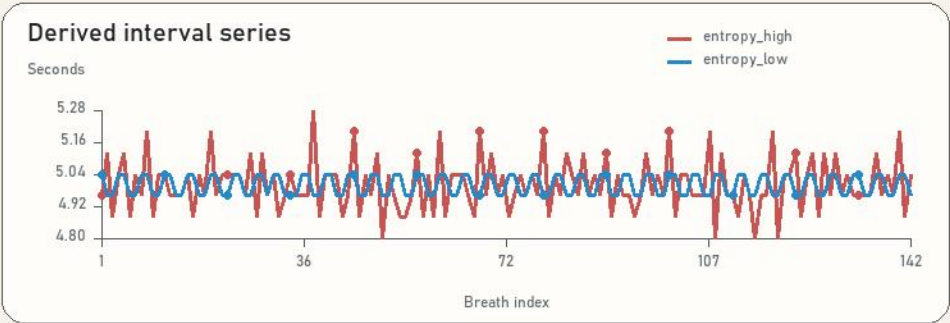
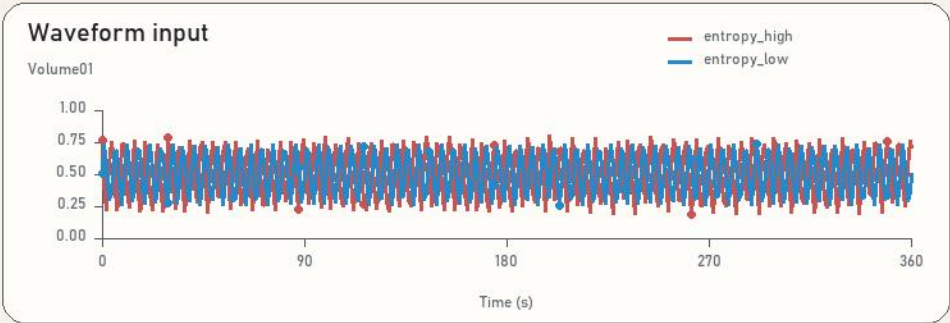


Time-domain summary				
Scenario	RMSSD	SDNN	pNN50	lnRMSSD
High	58.5	40.0	59.0	4.069
Low	3.0	4.1	0.0	1.108

The showcase preset uses a 120 s / 32 RR preview window so the examples stay deterministic and documentation-friendly.

Breathing-dynamics derivation

Breathing waveform samples become accepted extrema, interval/amplitude series, and then entropy-oriented feature bundles.



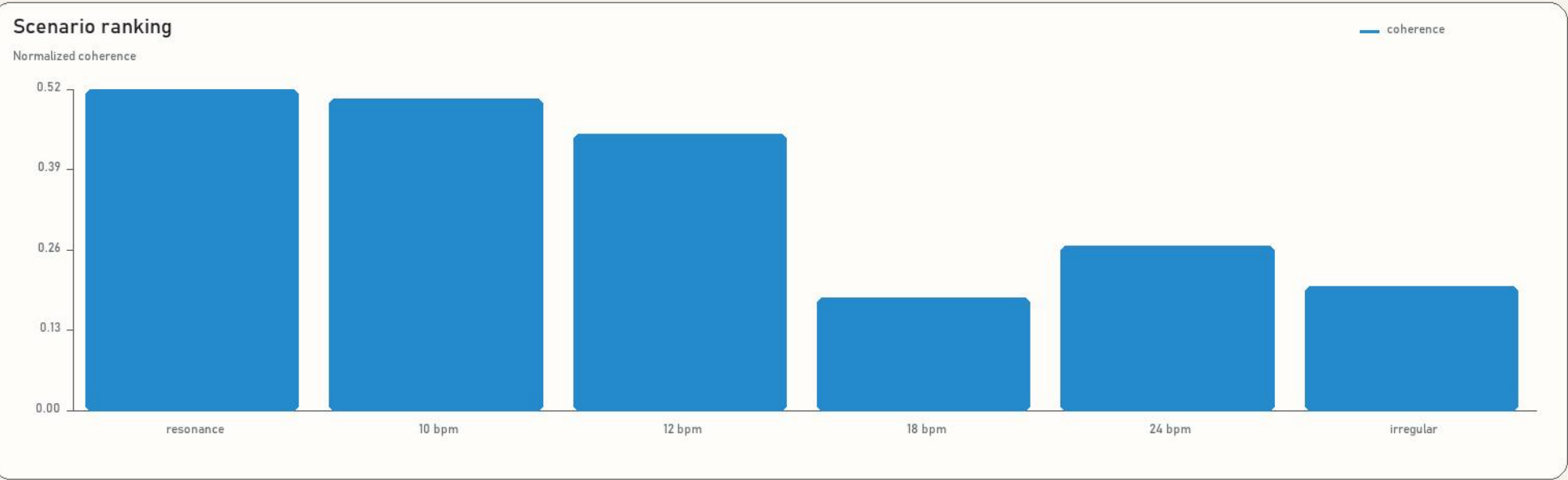
Entropy summary

Scenario	Int SampEn	Amp SampEn	Int CV	Amp CV
High	1.230	0.426	0.020	0.034
Low	0.044	0.148	0.008	0.002

The exported analysis includes the accepted extrema and derived series so supplementary figures can trace the metric path directly.

Coherence appendix

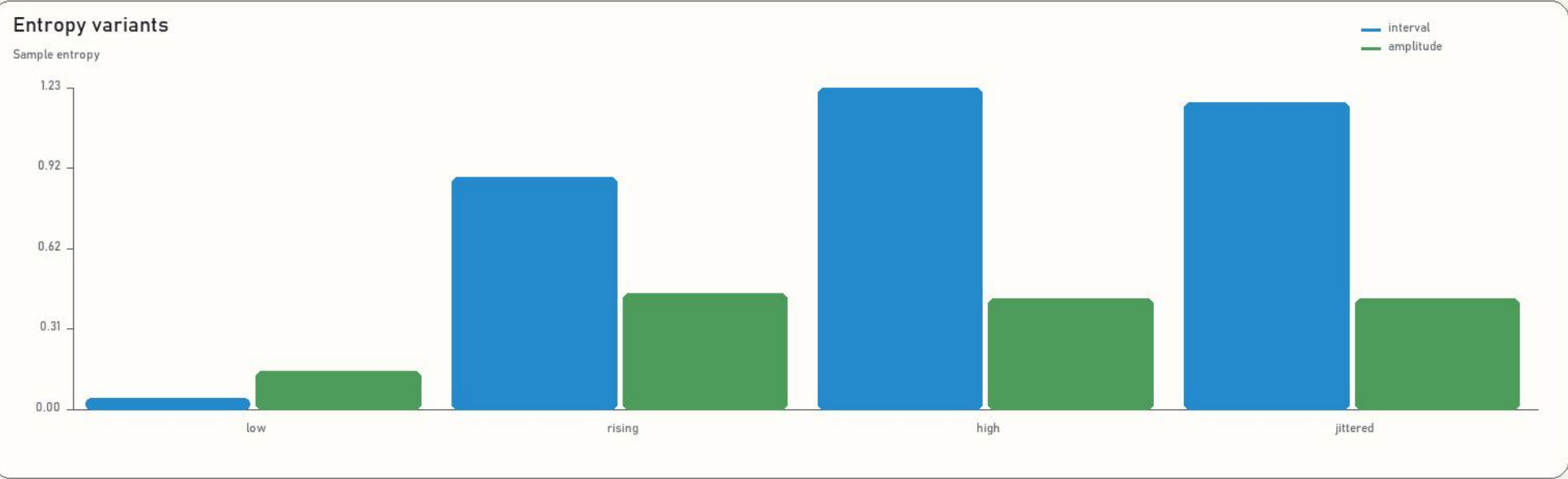
Resonance stays above the off-resonance sweep and irregular RR showcase.



Scenario values	
Scenario	Norm score
resonance	0.516
10 bpm	0.502
12 bpm	0.445
18 bpm	0.181
24 bpm	0.265
irregular	0.200

Dynamics appendix

Entropy variants and edge states stay visible for supplementary comparisons.



Edge-state checks

Scenario	State	Int SampEn	Amp SampEn
flat	2	--	--
pause	4	0.569	0.349